

Spotify Music Genre Classification

Austin Wu

April 27, 2025

1 Introduction and Methodology

This report presents a machine learning model for classifying Spotify songs into 10 genres based on audio features. The task involved addressing multiple challenges including missing values, non-normal distributions, and dimensionality reduction.

1.1 Addressing Data Challenges

The dataset presented several specific challenges that were addressed as follows:

- **Missing Values:** Negative duration values (-1.0) and other missing features were identified and replaced using KNN imputation with $k = 5$, which preserves feature relationships better than mean imputation or removal.
- **Non-normal Distributions:** Audio features showed significant non-normality, which was addressed through feature scaling and the use of tree-based models (XGBoost) that are robust to non-normal distributions.
- **Categorical Data:** The 'key' feature (string format) was one-hot encoded, while 'mode' (categorical) was binary encoded (Major=1, Minor=0) to convert them to useful numerical representations.
- **Multi-class Nature:** The 10-genre classification problem was approached through one-vs-rest classifiers and specialized binary classifiers for frequently confused pairs, with evaluation using macro-average AUC to account for all classes.
- **Dimensionality Reduction:** Categorical features were properly encoded before dimensionality reduction, ensuring mode and key were not inappropriately normalized during PCA.

1.2 Feature Engineering and Model Design

Based on musical domain knowledge and some research, I created some features capturing musical relationships:

- Basic relationship features like energy-loudness ratio and rhythm strength (danceability $\times$ energy)
- Genre-specific features targeting difficult distinctions (e.g., rap_vocal_focus for Hip-Hop/Rap distinction)
- PCA reduced the engineered feature space from 47 to 15 dimensions while preserving 95% of variance

XGBoost was specifically chosen as the primary classifier for several compelling reasons:

- **Robustness to non-normal distributions:** The audio features exhibited significant non-normality, and tree-based methods like XGBoost do not make distributional assumptions

- **Handling of missing values:** In addition to our imputation strategy, XGBoost’s built-in handling of sparse data provides additional robustness
- **Feature importance:** XGBoost also provides interpretable feature importance metrics, which is crucial for understanding which audio characteristics define genres
- **Performance with high-dimensional data:** XGBoost performs well with the complex feature interactions present in audio data
- **Regularization capabilities:** The L1 and L2 regularization in XGBoost helped prevent overfitting on engineered features

A hybrid ensemble approach was developed around XGBoost to maximize classification performance:

- XGBoost base classifier optimized via RandomizedSearchCV (max_depth=4, learning_rate=0.08, n_estimators=497)
- Specialized binary classifiers for commonly confused genre pairs to address multi-class complexity
- Genre-specific weighting to balance overall performance across the 10 genres

2 Results and Analysis

2.1 Performance and Feature Importance

The final model achieved a macro-average AUC of 0.9225. As shown in Figure 1, performance varied significantly across genres, with Classical (0.99) and Anime (0.97) showing highest separability while Alternative (0.84) proved most challenging.

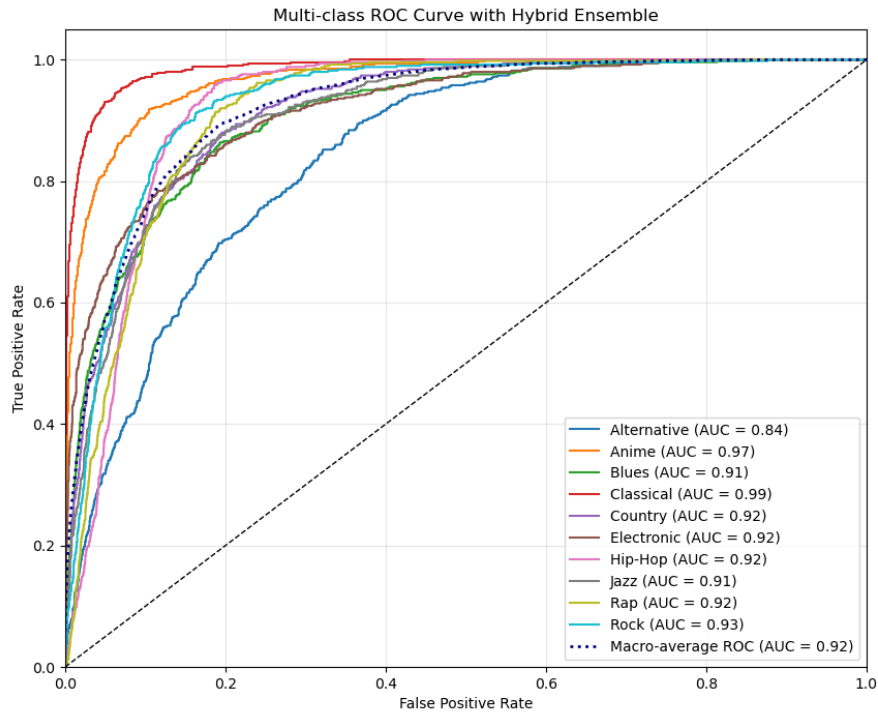


Figure 1: Multi-class ROC curves showing performance across all genres

2.2 Genre Clustering

The t-SNE visualization (Figure 2) reveals how genres cluster in reduced dimensional space:



Figure 2: t-SNE visualization showing genre clusters

Analysis of these clusters shows Classical music forming a distinct, isolated cluster (explaining its high AUC), while Hip-Hop and Rap substantially overlap. Alternative music appears widely dispersed across the space, explaining its lower classification accuracy. This clustering confirms that genre classification success directly corresponds to how acoustically distinct each genre is from others.

3 Conclusion

The most important factors underlying classification success were:

- Genre distinctiveness in audio feature space, with clearly separated genres (Classical, Anime) achieving highest AUC scores
- Targeted feature engineering for problematic genre pairs
- The acoustic-electronic spectrum as a primary organizing principle
- Effective dimensionality reduction preserving key variance

The final AUC of 0.9225 demonstrates that domain-specific feature engineering combined with specialized classification approaches can effectively address multi-class music genre classification, even with inherently fuzzy genre boundaries.

A Additional Visualizations and Analysis

A.1 Feature Distributions

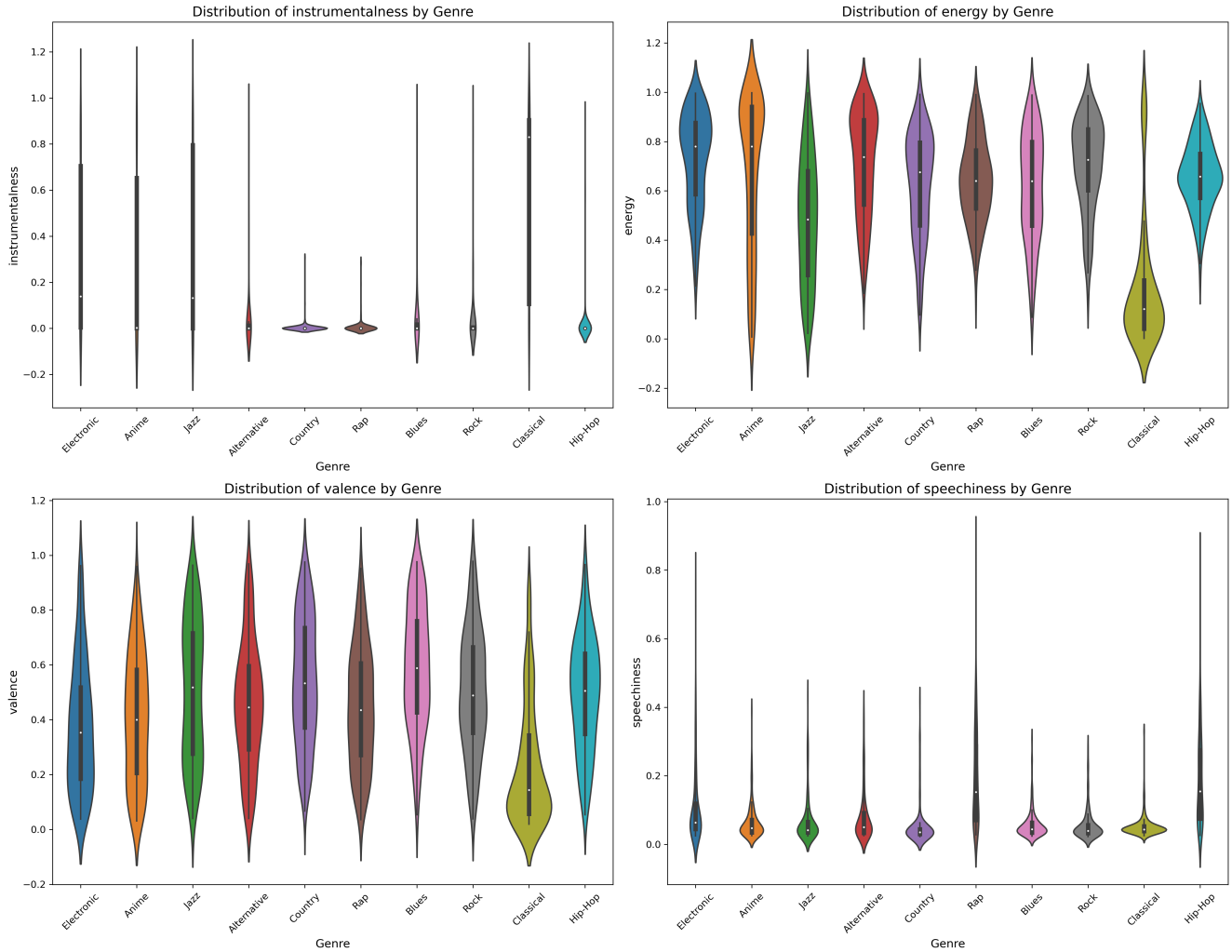


Figure 3: Violin plots of key audio features across genres. Note how instrumentalness clearly puts separates vocal-centric genres (Rap, Hip-Hop) from instrumental-heavy genres (Classical, Jazz, Electronic, and anime). Energy distributions show Classical music’s distinctively low energy profile compared to other genres. Valence (emotional positivity) distributions reveal genre-specific emotional tendencies, with Classical showing lower values, and Blues at surprising high values. Speechiness serves as a strong discriminator for Rap and Hip-Hop from other genres.

A.2 Genre Proximity Network

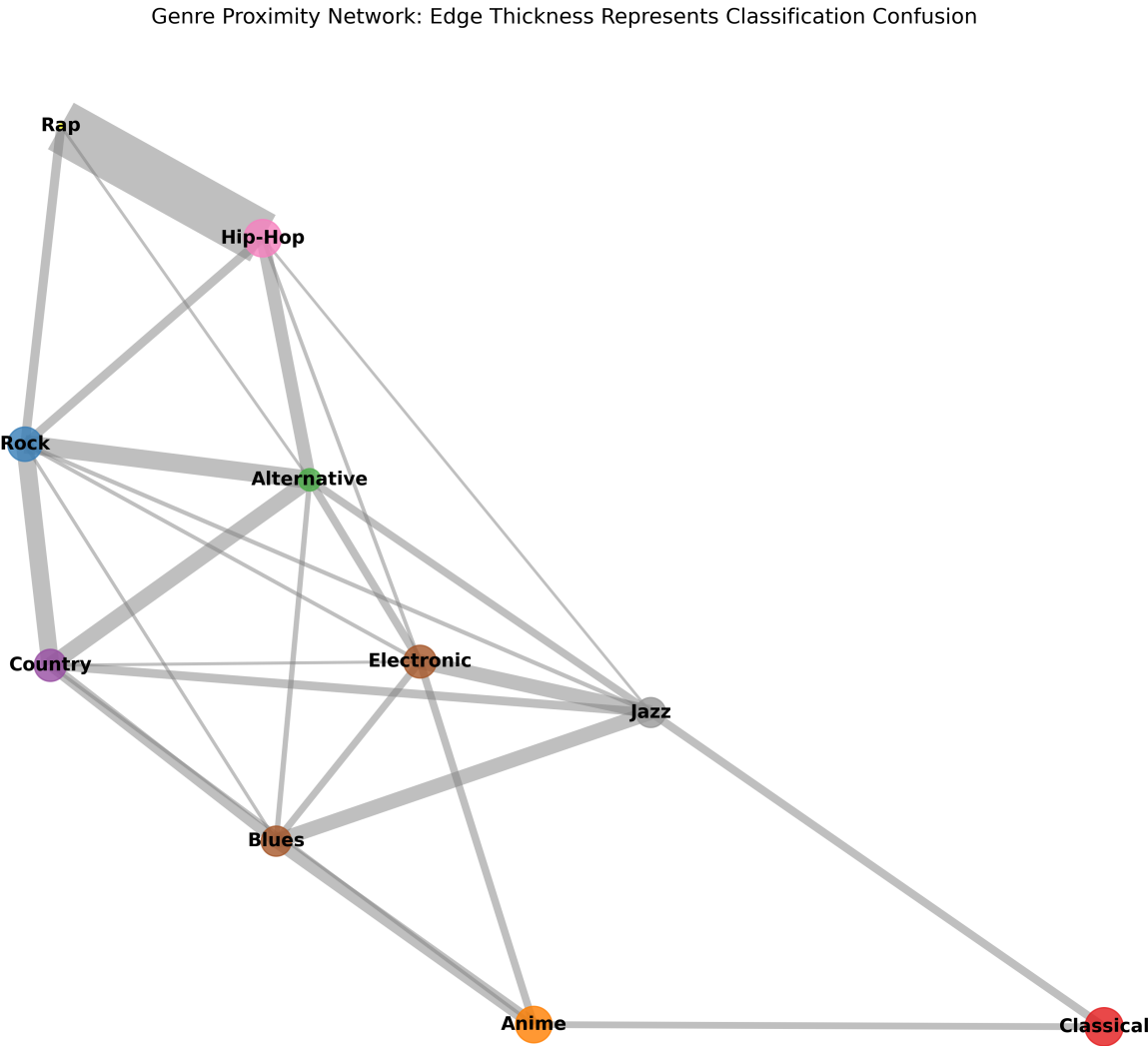


Figure 4: Network visualization of genre relationships based on classification confusion. Edge thickness represents the frequency of misclassification between genres. Three distinct clusters emerge: (1) Alternative-Rock-Country, (2) Hip-Hop-Rap, and (3) Blues-Jazz-Electronic. Classical and Anime form outlier nodes with minimal connections to other genres, explaining their high classification accuracy. Rock serves as a bridge genre with connections to multiple clusters, reflecting its influence across the musical spectrum.

A.3 Confusion Matrix

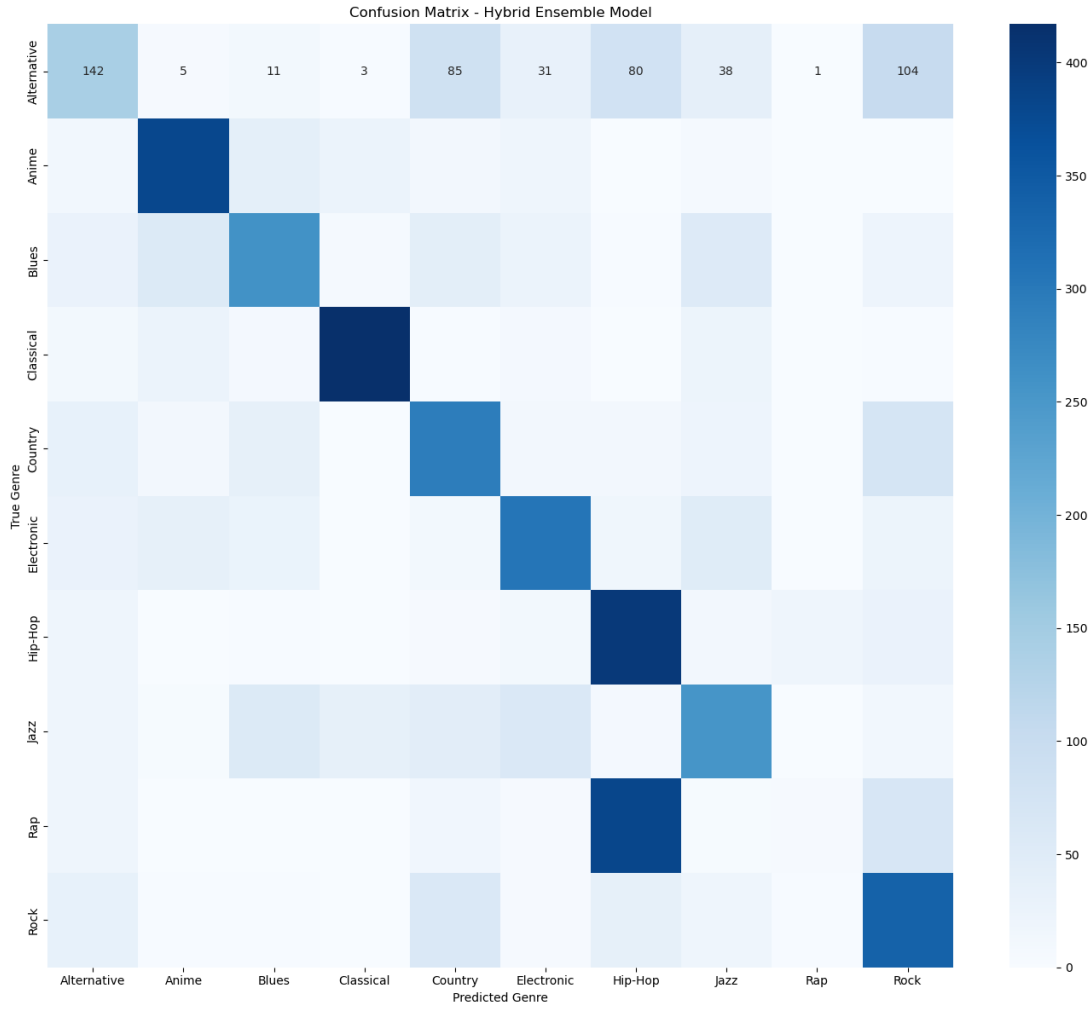


Figure 5: Confusion matrix showing classification results. The diagonal represents correct classifications, with brighter squares indicating higher accuracy. Key patterns include: Alternative music’s frequent confusion with Rock (91) and Country (89); significant mutual confusion between Hip-Hop and Rap despite specialized classifiers; Classical music’s exceptional accuracy with minimal misclassifications; and unexpected confusion between Electronic and Jazz (57), suggesting shared instrumental characteristics that weren’t fully captured by our engineered features.

A.4 Additional Insights and Future Work

- While our model achieved strong performance, linguistic information in artist and track names could potentially improve classification further
- The highly distributed nature of Alternative music suggests it might benefit from sub-genre classification
- Future work could explore deep learning approaches that might better capture complex audio patterns